روش‌های پیشرفته نمونه برداري در جنگل

استراتژي‌هاي نمونه‌برداري

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Stratified Random Sampling

In stratified sampling we use auxiliary information to stratify the entire population into, say, *H* strata.

Besides improving the variance efficiency of estimators other reasons to choose a stratified sampling design are

(1) estimates for homogeneous subpopulations (strata) may be required

(2) the desired precision is not the same for all subpopulation,

(3) assessment cost and/or attributes of interest are not the same for all subpopulations

(4) different sampling protocols apply to different subpopulations

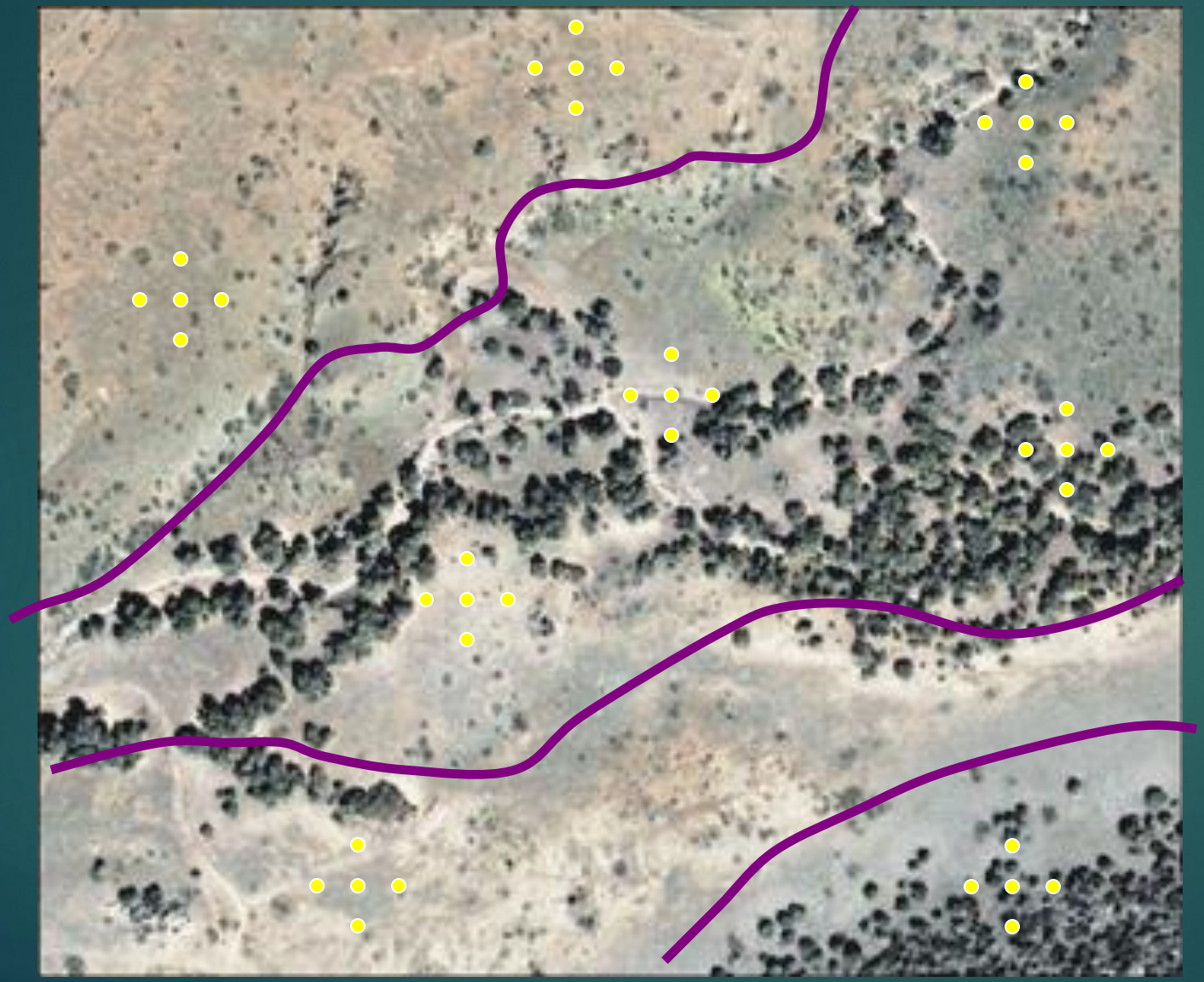
Stratified Random Sampling



Stratified Systematic Sampling



Stratified Cluster Sampling



Allocation methods

Equal allocation $$n_j = \frac{n}{M}$$

Proportional allocation $$n_j = \frac{N_j}{N} \times n$$

Neyman allocation $$n_j = \frac{N_j S_j}{\sum_{j=1}^{M} N_j S_j} \times n$$

Allocation methods

Optimal allocation

$$n_j = \frac{N_j \times \frac{S_j}{\sqrt{C_j}}}{\sum_{j=\backslash}^{M} (N_j \times \frac{S_j}{\sqrt{C_j}})} \times n$$

Estimators

$$\bar{x}_j = \frac{\sum_{i=1}^{n} x_i}{n_j}$$

$$\bar{x} = \sum_{j=1}^{M} \frac{N_j}{N} \times \bar{x}_j$$

$$SE_j = \sqrt{\frac{S_j^2}{n_j}}$$

$$SE = \sqrt{\sum_{j=1}^{M} [(\frac{N_j}{N})^2 \times SE_j^2]}$$

Poststratification

If, the population ratio Nj /N is known for each of the L strata, but it may not be feasible to draw the sample before conducting the field survey. In such cases, it is more efficient to poststratify the population, i.e., to allocate the sampling units on the basis of quantitative or qualitative information obtained from the sample.

$$s^2_{\bar{y}_{\text{p.str}}} = \frac{N-n}{n \cdot N} \sum_{J=1}^{L} \left(\frac{N_j}{N} \cdot \sigma_j^2 \right) + \frac{1}{n^2} \sum_{j=1}^{L} \left(\frac{N - N_j}{N} \cdot \sigma_j^2 \right)$$

Some points on stratification

- The effect of inaccurate determination of the sizes of the strata, Dated aerial photographs make estimates of the sizes of the strata imprecise.

- While stratification for the estimation of a single population attribute is generally advantageous, the situation is less clear for a multipurpose inventory with its many attributes of interest.

- If the population attributes of interest have changed between two inventories the stratification used in the older inventory may be inopportune.

Multiphase sampling

- Two phase sampling with regression estimator
- Two phase sampling with ratio estimator
- Two phase sampling for stratification

Two phase sampling with regression estimator

In the first phase a large sample of size n_1 is selected;
in the second phase a random subsample of size $n_2 \leq n_1$ is selected where the auxiliary variable, x, and the variable of interest, say y, are measured.

The two-phase sample with regression estimator of the population mean for a single variable of interest and a single auxiliary variable is:

$$\hat{\bar{Y}}_{2prgr} = \hat{\bar{Y}}_2 + \hat{\beta} \times \left(\hat{\bar{X}}_1 - \hat{\bar{X}}_2 \right)$$

where $\hat{\bar{X}}_1$ and $\hat{\bar{X}}_2$ are the sample-based estimates of the mean of x in the first-phase sample and in the second-phase sample, respectively, $\hat{\bar{Y}}_2$ is the estimate of the population mean of y obtained from the second-phase sample, and $\hat{\beta}$ is the least-squares regression coefficient of y on x computed from the second-phase sample. Note, $\hat{\beta}$ can be improved by recognizing that the first-phase

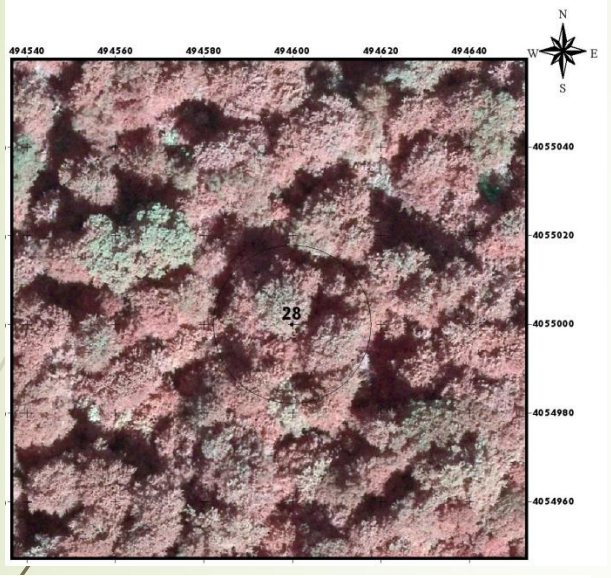
Two phase sampling with regression estimator

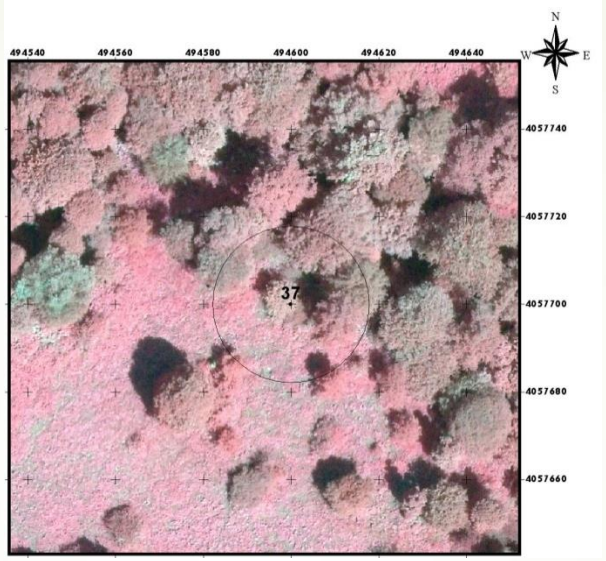
An estimator of the variance of $\hat{\bar{Y}}_{2\text{prgr}}$ is

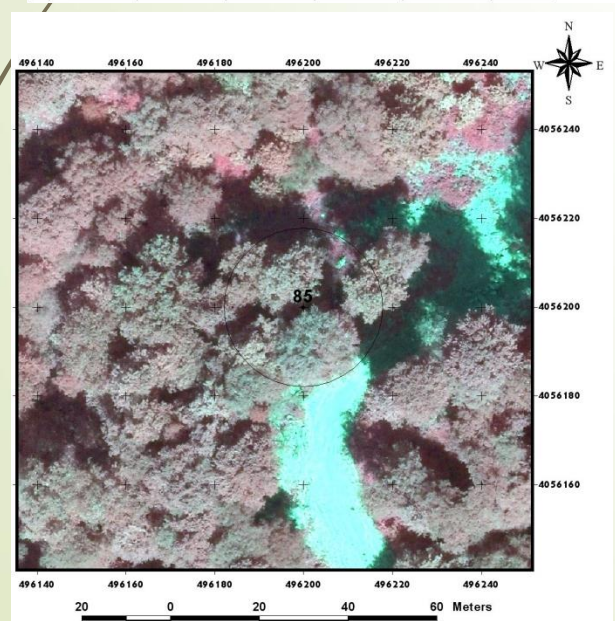
$$\hat{\text{var}}\left(\hat{\bar{Y}}_{2\text{prgr}}\right) = \hat{\text{var}}\left(y|x\right) \times \left[\frac{1}{n_2} + \frac{\left(\hat{\bar{X}}_2 - \hat{\bar{X}}_1\right)^2}{\sum_{j=1}^{n_2}\left(x_j - \hat{\bar{X}}_2\right)^2}\right] + \frac{\hat{\text{var}}\left(y\right) - \hat{\text{var}}\left(y|x\right)}{n_1} - \frac{\hat{\text{var}}\left(y\right)}{N}$$

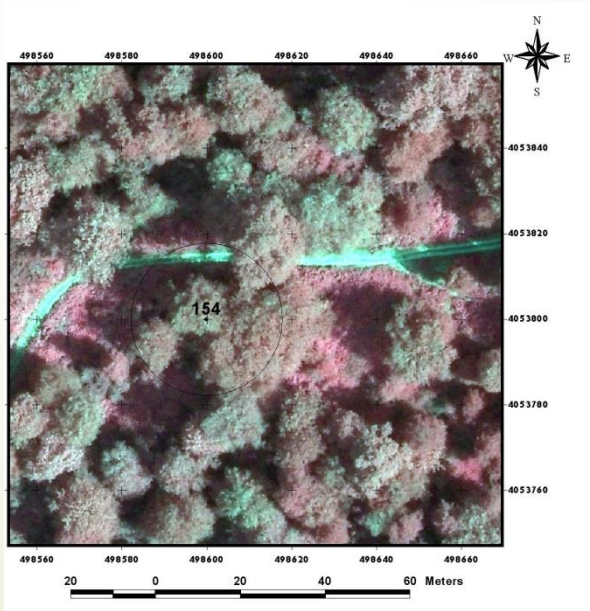
where $\hat{\text{var}}\left(y|x\right)$ is the conditional variance of y given x, or simply the variance of the expected values of y once x is known. For a linear relationship $y=\alpha+\beta x+\text{error}$, the conditional variance of y is the variance of the linear prediction $\alpha+\beta x$. The first term on the right-hand side of this variance expression contains the variance of the linear predictions of y, while the second term adds the variance of the prediction errors. The third term is a correction factor for finite-population predictions. In infinite populations the last term drops out.

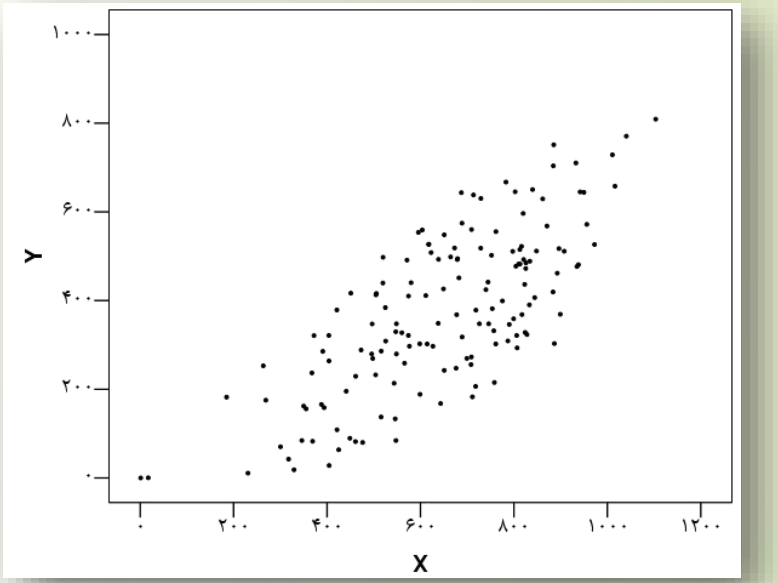
ارزیابی هزینه و دقت نمونه برداری دوگانه رگرسیون مبنا برای برآورد موجودی حجمی جنگل با استفاده از تصاویر هوایی











Two phase sampling with ratio estimator

The ratio estimator assumes that the population total X for the auxiliary variable x is known. In double sampling for ratio estimation, the population mean X is estimated from the initial sample of size n_1. A subsample n_2 is drawn to calculate the ratio estimator R_{rm}. The ratio estimate for the population mean of y is

$$\bar{y}_{ds_{rm}} = \widehat{R}_{rm} \cdot \bar{x}_1 = \frac{\bar{y}}{\bar{x}_2} \cdot \bar{x}_1$$

with variance

$$s^2_{\bar{y}_{\mathrm{ds}rm}} = \frac{s_y^2 - 2 \cdot R_{rm} \cdot s_{xy} + R_{rm}^2 \cdot s_x^2}{n_2} + \frac{2 \cdot R_{rm} \cdot s_{xy} + R_{rm}^2 \cdot s_x^2}{n_1} - \frac{s_y^2}{N}$$

Two phase sampling for stratification

Two-phase sampling for stratification is similar to stratified sampling except for the fact that the sizes of the strata are not known without error. Strata sizes are estimated by the larger, first-phase sample and the variable/attribute of interest is assessed in the second phase.

In two-phase sampling for stratification an auxiliary categorical variable that can take one of H distinct values is sampled in the first phase for the purpose of estimating the proportion of the population elements/units in each category (stratum).

The second-phase sample can be a subsample of units sampled from the first phase

Two phase sampling for stratification

estimator for the population mean

$$\hat{\bar{Y}}_{2pstr} = \sum_{h=1}^{H} \frac{n_{1h}}{n_1} \times \hat{\bar{Y}}_h = \sum_{h=1}^{H} \hat{W}_h \times \hat{\bar{Y}}_h,$$

where n_{1h} is the first-phase sample size in stratum h (h=1, . . . , H), $n_i = n_{i1} + n_{i2} + \ldots + n_{iH}, i = 1,2$, and $\hat{\bar{Y}}_h$ is the second-phase estimate of the mean of the hth stratum.

estimator of the variance

$$\hat{\text{var}}\left(\hat{\bar{Y}}_{2pstr}\right) = \frac{N-1}{N} \sum_{h-1}^{H} \left(\frac{n_{1h}}{n_1 - 1} - \frac{n_{2h}-1}{N-1} \right) \frac{W_h \hat{\text{var}}\left(y_{2h}\right)}{n_{2h}} + \frac{N-n_1}{N\left(n_1-1\right)} \sum_{h=1}^{H} W_h \left(\bar{Y}_{2h} - \hat{\bar{Y}}_{2pstr} \right)^2$$

where $\hat{\text{var}}\left(y_{2h}\right)$ is a second-phase sample-based estimate of the variance of variable y in the hth stratum.

The advantage of double sampling for stratification

The advantage of double sampling for stratification over stratified sampling is that a laborious assessment of the sizes of the strata is replaced by a quicker and less costly sampling procedure (Sutter 1990).

In comparison with double sampling with regression estimators:

The derivation of regression functions has been eliminated. In many practical situations there is no suitable auxiliary variable that is uniformly and strongly correlated with the variable of interest across the entire population.

This argument extends to the case where several variables of interest are to be determined and where field samples cannot be located with any great precision or linked with a desired accuracy to observations made in the first phase.

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